# Project Nema
## An Automated Aquarium

Project Report

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## Abstract

This project report presents the development of an innovative automated aquarium system leveraging Arduino technology. The primary objective is to create a comprehensive ecosystem within the aquarium through the integration of various sensors and actuators. The system incorporates pH sensors, a water level indicator, temperature control mechanisms, a GSM module for remote monitoring, and a servo motor for precise and automated feeding. The implementation allows for real-time monitoring and control, ensuring optimal environmental conditions for aquatic life while enabling remote accessibility for seamless management.


### Key Points

| | | |
|---|---|---|
| **pH Sensors** | **Water Level Indicator** | **Temperature Control** |
| **GSM Module** | **Servo Motor / Auto Feeding** | **Arduino Automation** |
| **Real-Time Monitoring** | **Remote Accessibility** | **IoT Implementation** |

# 1. Introduction

### 1.1 Purpose

In the realm of aquatic habitat maintenance, the amalgamation of technology and ecological preservation has taken a pioneering leap. This project endeavors to redefine the paradigm of aquarium management, where precision meets environmental sensitivity through the fusion of cutting-edge technology and intricate ecological understanding.

### 1.2 Objectives

- **Ecosystem Harmony:** Forge an automated system that actively sustains a harmonious aquatic ecosystem by regulating critical environmental parameters such as pH levels and temperature.
- **Technical Ingenuity:** Confront the complexities of integrating diverse technological components — GSM modules, servo motors, and precise water-level control mechanisms.
- **Environmental Precision:** Attain a level of precision that caters to the nuanced needs of aquatic life, ensuring a stable and conducive environment for their well-being and longevity.

- **Symbiosis of Automation and Ecology:** Bridge the gap between technological automation and ecological sensitivity, demonstrating how sophisticated systems can coexist with the natural environment within an aquarium.

# 2. Overview of Components

## 2.1 Principal Components

### 2.1.1 Arduino Uno



*Fig 2.1 — Arduino Uno (central microcontroller)*

The Arduino Uno acts as the central intelligence in the automated aquarium project, orchestrating sensor data interpretation and precise control of actuators. It manages pH sensors, water level indicators, temperature controls, a GSM module for remote monitoring, and servo motors for automated feeding. Its scalability allows seamless integration of new sensors or features.

### 2.1.2 GSM Module (SIM800L)



*Fig 2.2 — SIM800L GSM Module*

Integrated within the aquarium setup, the GSM module enables instant communication with the user. Coupled with the Arduino Uno, this module transmits real-time updates and notifications directly to the user's mobile device. In case of any deviations in vital parameters (pH or temperature beyond set thresholds), the GSM module triggers immediate SMS alerts.

### 2.1.3 Temperature Sensor

The temperature sensor serves as the vigilant guardian of thermal stability. Linked to the Arduino Uno, it continuously monitors water temperature with precision, ensuring the aquatic environment remains within the predefined range vital for the well-being of aquatic life. When temperature drops below the threshold, it actively triggers heater activation.

### 2.1.4 pH Sensor

At the core of the system's chemical equilibrium lies the pH sensor, functioning as the custodian of water quality. It vigilantly scrutinizes pH levels — a critical factor in sustaining a healthy aquatic environment. In the event of pH fluctuations beyond optimal ranges (pH > 10), the sensor triggers the GSM module to send an immediate alert.

### 2.1.5 Water Level Indicator (SONAR / HC-SR04)



*Fig 2.3 — HC-SR04 Ultrasonic Sensor (SONAR)*

The SONAR-based water level indicator uses ultrasonic waves to precisely measure water levels. Integrated with the Arduino Uno, it ensures the aquarium maintains an ideal water volume, triggering necessary pump actions to maintain stability.

### 2.1.6 Heater

Paired with the Arduino Uno, the heater uses precise temperature control for maintaining optimal water conditions. When temperature falls below 25°C, the Arduino activates the heater via the digital output pin, with an LED indicator illuminating simultaneously to signal activation.

### 2.1.7 Water Pump



*Fig 2.4 — Water Pump*

Integrated with the Arduino Uno, the water pump facilitates periodic water changes, enhancing circulation and maintaining water quality. Both inlet and outlet pumps are controlled by Arduino digital pins, and can be triggered remotely via a GSM command ('m').

### 2.1.8 Servo Motor



*Fig 2.5 — Servo Motor (fish feeder mechanism)*

Integrated with the Arduino Uno, the servo motor enables precise and automated fish feeding by controlling the dispensing mechanism. After a 12-hour interval, the Arduino triggers the servo motor to rotate to 45° (dispensing food) then returns to 0°, ensuring consistent scheduled nourishment.

## 2.2 Basic Components

| Component | Component | Component |
|---|---|---|
| Vero Board | Bread Board | Resistor |
| MOSFET | OLED Display | Jumper Wire |
| LED Bulb | Pipes | Demo Aquarium |

# 3. System Operation Overview

## 3.1 Methodology Flow

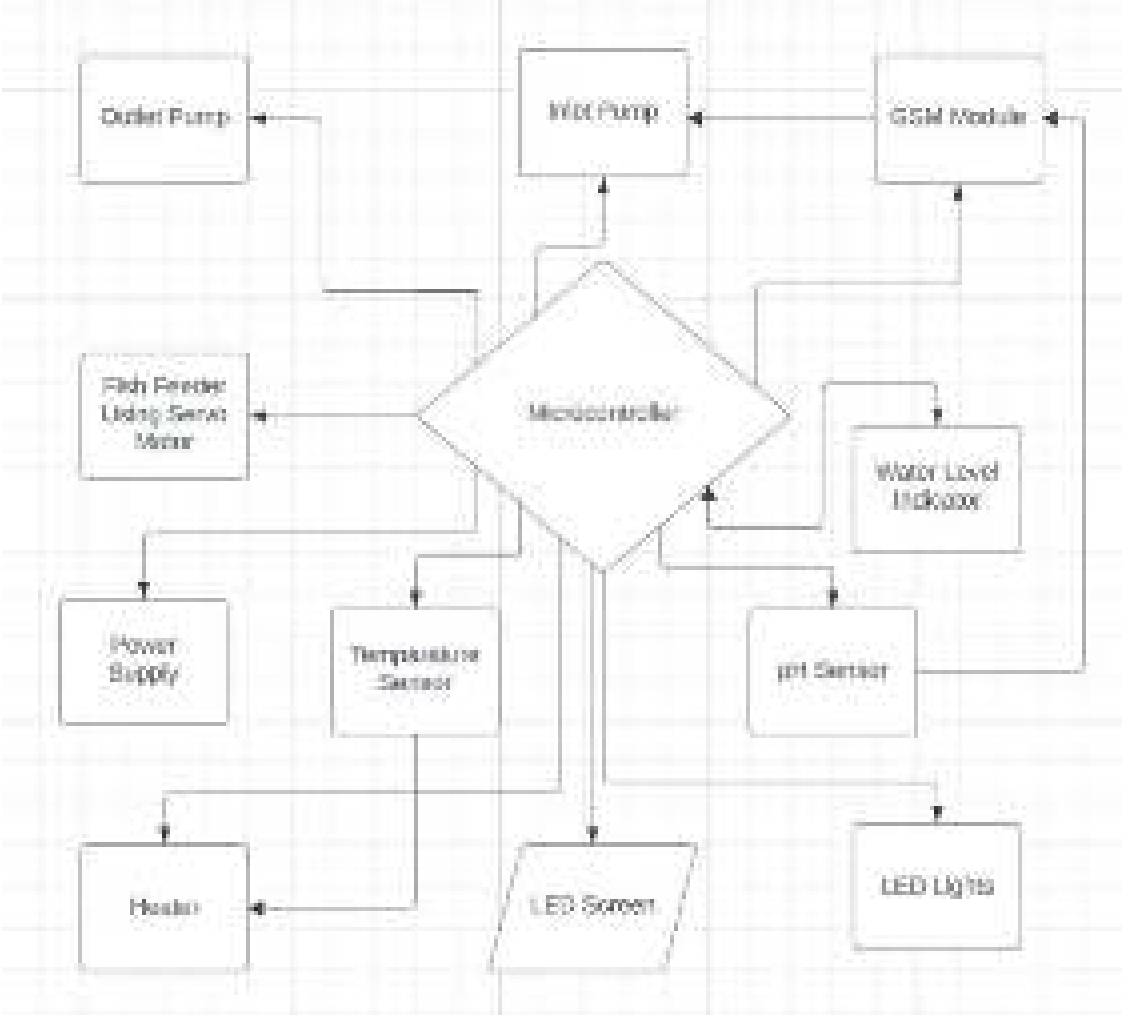



*Fig 3.1 — System methodology flow diagram*

The microcontroller (Arduino Uno) sits at the center of all operations, interfacing bidirectionally with every module. Sensors feed data in; the controller decides what actuators to trigger; users interact remotely via the GSM module.

## 3.2 Circuit Diagrams

### 3.2.1 GSM Module — Arduino Connection

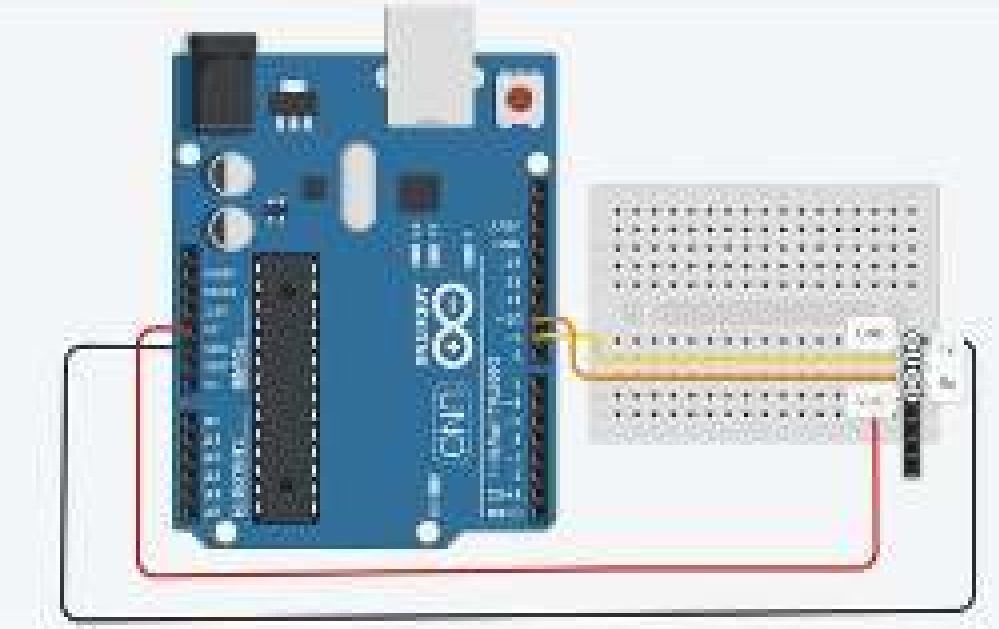


*Fig 3.2 — GSM module wiring to Arduino Uno*

### 3.2.2 Temperature Sensor — Arduino Connection

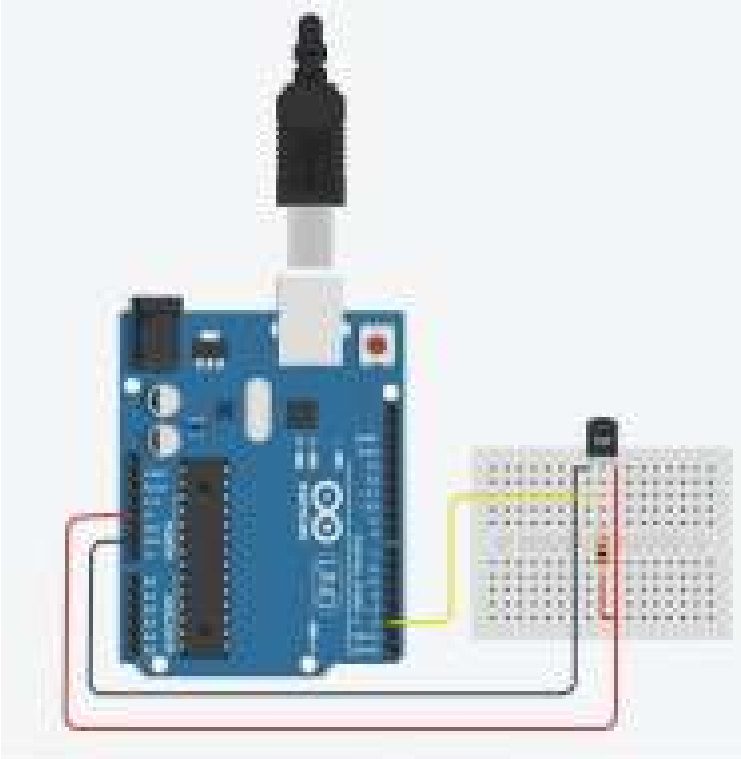


*Fig 3.3 — Temperature sensor wiring to Arduino Uno*

### 3.2.3 SONAR — Arduino Connection

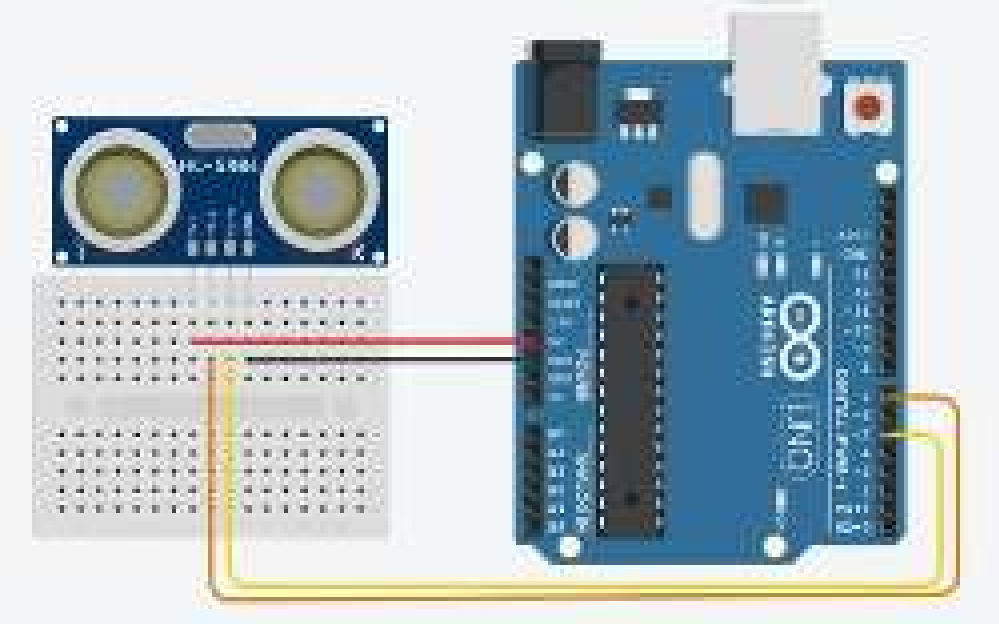


*Fig 3.4 — HC-SR04 SONAR wiring to Arduino Uno*

### 3.2.4 pH Sensor — Arduino Connection

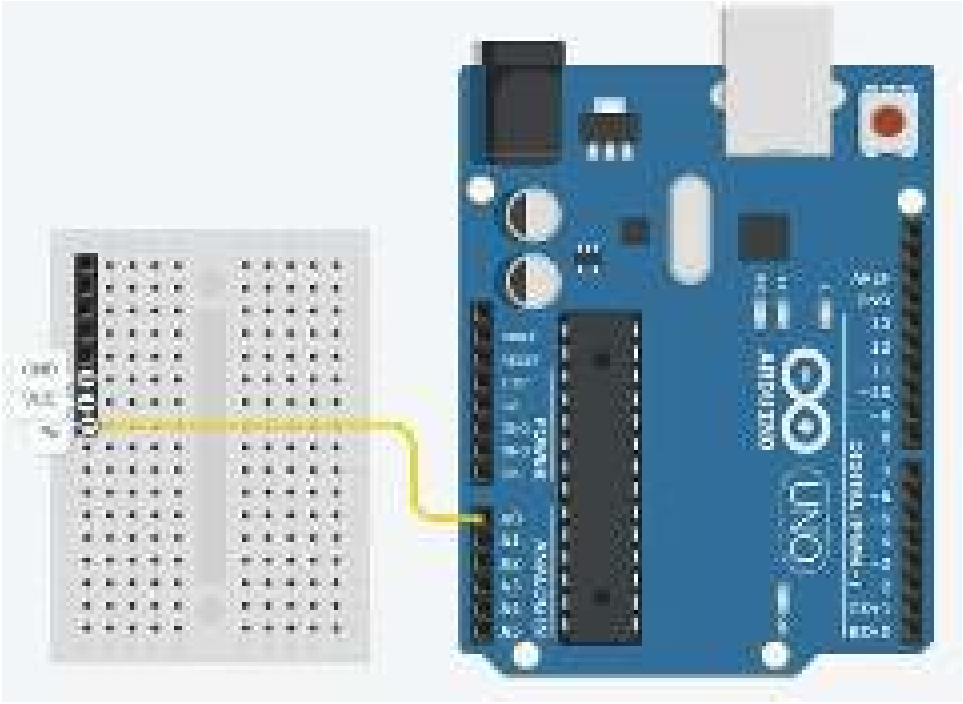


*Fig 3.5 — pH sensor wiring to Arduino Uno*

## 3.3 Functionality Analysis

### 3.3.1 Sensor Data Display on OLED

The system gathers sensor readings — pH, temperature, and water level — and displays them on an OLED screen in real time. This allows users to monitor crucial aquarium parameters conveniently and promptly.

### 3.3.2 pH Sensor & GSM Integration

**Trigger condition:** pH > 10. When detected, the GSM module sends an immediate SMS alert to the registered mobile number. The alert message reads: **"Boss I'm dying."** The boolean flag **msg_sent_ph** prevents duplicate messages until pH returns to safe levels (≤ 5).

### 3.3.3 User-Initiated Water Change via GSM & Pump Control

Upon receiving the character **'m'** via SMS, the Arduino activates motor1 (inlet pump) to HIGH, initiating a water change. The outlet pump is activated automatically when the SONAR detects that water distance ≥ 12 cm (low level).

### 3.3.4 Heater Activation & Temperature Monitoring

**Trigger condition:** Water temperature < 25°C. The Arduino sets TEMP_ALERT_PIN (pin 8) HIGH, activating the heater and simultaneously illuminating the LED indicator. When temperature returns to ≥ 25°C, the pin is set LOW.

### 3.3.5 Automated Feeding with Servo Motor

After every **12-hour interval** (43,200,000 ms delay), the Arduino triggers the servo motor to rotate to **45°** (dispensing food), then returns to **0°** after 1 second. This provides consistent scheduled nourishment without manual intervention.

# 4. Features

## 4.1 Automation Highlights

- **Intelligent Sensor Integration:** pH, temperature, and water-level sensors feed real-time data to the Arduino, which uses conditional thresholds to trigger actions — heater, pumps, or GSM alerts — automatically.
- **Precise Servo Motor Control:** Timed loops within the Arduino code activate the servo motor at predefined 12-hour intervals, dispensing fish feed with precision.

## 4.2 Advanced Monitoring Techniques

- **Real-Time OLED Visualization:** Sensor data is formatted and pushed to the OLED display every loop cycle, providing an instant visual readout of pH, water level, and temperature.

- **GSM Remote Monitoring & Alerts:** Conditional statements in the Arduino code detect threshold breaches and dispatch SMS notifications via the GSM library, ensuring the user is informed regardless of physical proximity.

## 4.3 Algorithmic Control Precision

- **Conditional Logic:** if-else statements interpret sensor readings against thresholds, triggering heater activation, pump operation, or GSM alerts as required.
- **Timed Sequences:** delay() and loop timing functions govern the 12-hour feeding schedule, ensuring predictable and consistent automation.

# 5. Implementation

## 5.1 Hardware Implementation

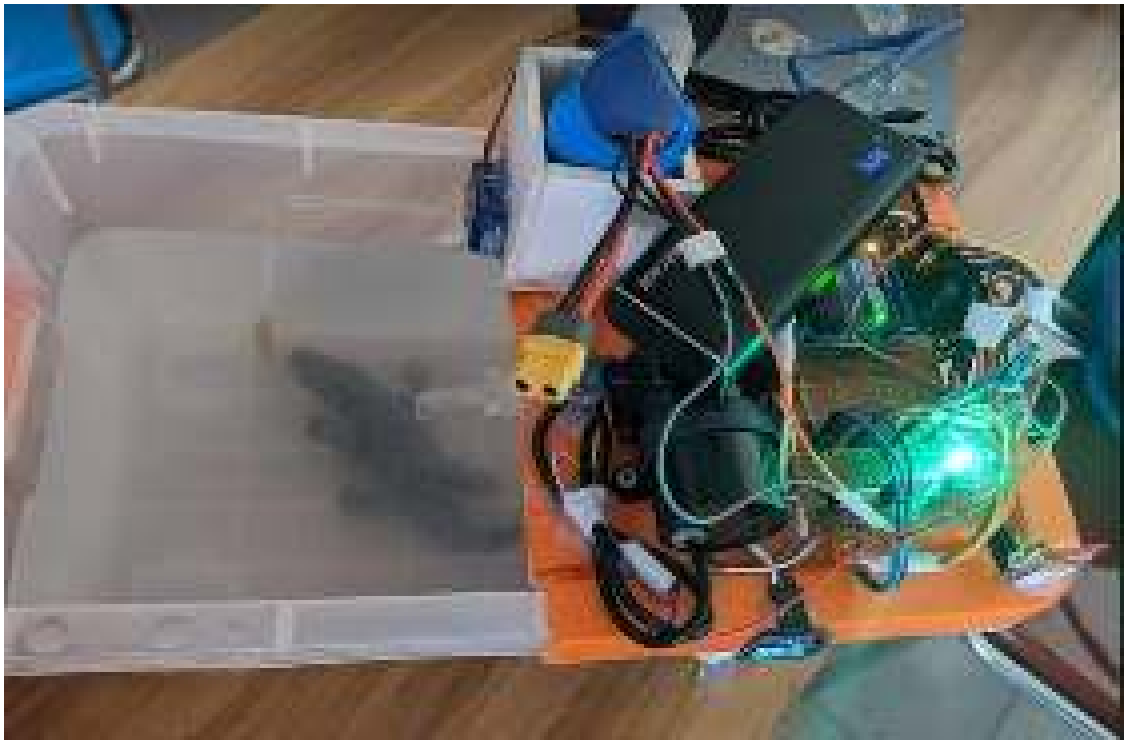


*Fig 5.1 — Complete assembled automated aquarium system*

### 5.2.1 Sensors Integration

- **pH Sensor:** Connected to Arduino analog pin A0 with voltage divider. Outputs voltage proportional to solution acidity, interpreted by the calibration formula: pH = −5.70 × V + 34.50
- **Temperature Sensor:** Interfaced via digital pin 2 using OneWire protocol (DallasTemperature library). Transmits digital temperature values.
- **Water Level (SONAR):** Trigger on digital pin 7, Echo on digital pin 5. Uses pulseIn() to measure ultrasonic round-trip time and compute distance in cm.

### 5.2.2 Actuators and Modules

- **Heater:** Connected to digital pin 8 via relay/transistor. Activated HIGH when temperature < 25°C.
- **Servo Motor:** Connected to PWM pin 11. Moves to 45° to dispense feed, returns to 0° after dispensing.
- **GSM Module:** Software serial on pins 9 (TX) and 10 (RX). AT command protocol for SMS send/receive.
- **OLED Display:** I2C interface on analog pins A4 (SDA) and A5 (SCL). Uses Adafruit_SSD1306 library at address 0x3C.
- **Water Pump:** Motor1 on digital pin 6, Motor2 on digital pin 3. Activated based on SONAR distance readings and remote GSM commands.

## 5.3 Arduino Pin Connections

### 5.3.1 SONAR (HC-SR04) → Arduino

| SONAR Pin | Arduino Pin |
|---|---|
| GND | GND |
| Echo | Digital 5 |
| Trig | Digital 7 |
| Vcc | 5V |

### 5.3.2 pH Sensor → Arduino

| Sensor Pin | Arduino Pin |
|---|---|
| V+ | 5V |
| GND | GND |
| P0 | Analog A0 |

### 5.3.3 OLED Display → Arduino

| OLED Pin | Arduino Pin |
|---|---|
| GND | GND |
| Vcc | 5V |
| SCL | Analog A5 |
| SDA | Analog A4 |

### 5.3.4 Temperature Sensor → Arduino

| Sensor Wire | Arduino Pin |
|---|---|
| Yellow (Data) | Digital 2 |
| Red | 5V |
| Black | GND |

### 5.3.5 GSM Module → Arduino

| GSM Pin | Arduino Pin |
|---|---|
| Vcc | 5V |
| GND | GND |

| | |
|---|---|
| Tx | Digital 9 |
| Rx | Digital 10 |

### 5.3.6 Servo Motor → Arduino

| Motor Pin | Arduino Pin |
|---|---|
| Signal | Digital 11 (PWM) |
| Vcc | 5V |
| GND | GND |

# 6. Evaluation of the Automated Aquarium System

## 6.1 Performance Analysis

The performance analysis reveals commendable reliability in maintaining the aquarium's environmental parameters. Critical evaluation highlights occasional latency in sensor responsiveness during rapid environmental fluctuations. Minor inaccuracies in certain sensor readings demand fine-tuning or calibration to optimize system responsiveness and accuracy.

## 6.2 Impact

The system has significantly reduced manual workload associated with monitoring and adjusting aquarium parameters. Automated functionalities have led to a discernible improvement in water quality, promoting a healthier ecosystem. GSM integration empowers users with timely alerts, enabling swift intervention to preserve aquarium stability.

## 6.3 User Experience

Users appreciate the intuitive OLED display providing real-time insights into crucial parameters. Automated feeding via the servo motor has received overwhelmingly positive feedback. The GSM remote accessibility has been lauded, though the interface's simplicity warrants enhancement. There is a growing desire for an expanded user interface allowing greater customization.

# 7. Future Enhancements

## 7.1 Integration of Oxygen Sensor

- Addition of a dissolved oxygen sensor for real-time monitoring of $O_2$ levels.
- Enables precise adjustments to maintain optimal oxygen levels crucial for aquatic life.
- Strengthens the system's capability to sustain a healthy aquatic environment.

## 7.2 Implementation of AI Technology

- Incorporation of AI for predictive analysis and adaptive control.
- AI-driven learning from data patterns to predict changes in aquarium parameters.
- Facilitates preemptive adjustments, optimizing the environment proactively.
- Evolves the system into an intelligent, self-regulating platform for enhanced management.

## 8. Challenges

| Challenge | Description |
|---|---|
| **Synchronization Delays** | Delays between sensor readings and actuator responses caused intermittent disruptions, demanding fine-tuning for real-time responsiveness. |
| **GSM Module Coding** | Ensuring seamless data transmission and reception required meticulous AT command development and testing for reliable remote communication. |
| **Water Level Sensor Issues** | Initial setbacks with traditional water level sensors prompted adoption of SONAR technology for accurate and consistent measurements. |
| **Code Optimization** | Sluggish code execution led to system inefficiencies; dedicated efforts were invested in optimizing code for faster and streamlined performance. |
| **Code Syntax Errors** | Frequent syntax errors during development demanded meticulous debugging and thorough validation to ensure smooth execution. |
| **Module Integration Debugging** | Integrating diverse modules demanded rigorous debugging to harmonize functionalities and ensure seamless coordinated operation. |

## 9. Estimated Cost

| Item | Cost (BDT) |
|---|---|
| Push Button | 20 |
| Arduino Uno R3 | 1,100 |
| Vero Board | 30 |
| Resistor | 10 |
| Bread Board | 160 |
| OLED Display | 700 |
| Jumper Wire | 100 |
| GSM Module | 2,500 |
| Servo Motor | 1,500 |
| Pump | 450 |

| | |
|---|---|
| Heater | 500 |
| Temperature Sensor | 800 |
| pH Sensor | 2,500 |
| **TOTAL** | **10,370 BDT** |

# 10. Conclusion

As this aquarium project concludes, it has been an exhilarating journey merging cutting-edge technology with the marvels of nature. Sensors and Arduinos work together underwater, orchestrating a synchronized ballet of data and commands. The project demonstrates how sophisticated IoT systems can intricately coexist with, and enhance, the natural environment within an aquarium setting.

Fish tanks are no longer merely glass boxes — they are intricate ecosystems designed for the ultimate comfort and joy of aquatic life. This fusion of tech and nature, where innovation and nature's brilliance collide, paves the way for the most extraordinary automated aquatic realms.